

RMO(INDIA) – 2012(3)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Let ABCD be a unit square. Draw a quadrant of a circle with A as centre and B, D as end points of the arc. Similarly draw a quadrant of a circle with B as centre and A, C as end points of the arc. Inscribe a circle Γ touching the arc AC internally, the arc BD internally and also touching the side AB. Find the radius of the circle Γ .
 2. Let a, b and c are positive integers such that a divides b^5 , b divides c^5 and c divides a^5 . Prove that abc divides $(a + b + c)^{31}$.
 3. Let a and b are positive real numbers such that $a + b = 1$. Prove that $a^a b^b + a^b b^a \leq 1$.
 4. Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find the number of pairs $\{A, B\}$ such that $A \subseteq B, B \subseteq X, A \neq B$ and $A \cap B = \{5, 7, 8\}$.
 5. Let ABC be a triangle. Let D, E be a points on the segment BC such that $BD = DE = EC$. Let F be the mid point of AC. Let BF intersect AD in P and AE in Q respectively. Determine the ratio of the area of the triangle APQ to that of the quadrilateral PDEQ.
 6. Find all positive integers n such that $3^{2n} + 3n^2 + 7$ is a perfect square.

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